

UNSUPERVISED LEARNING

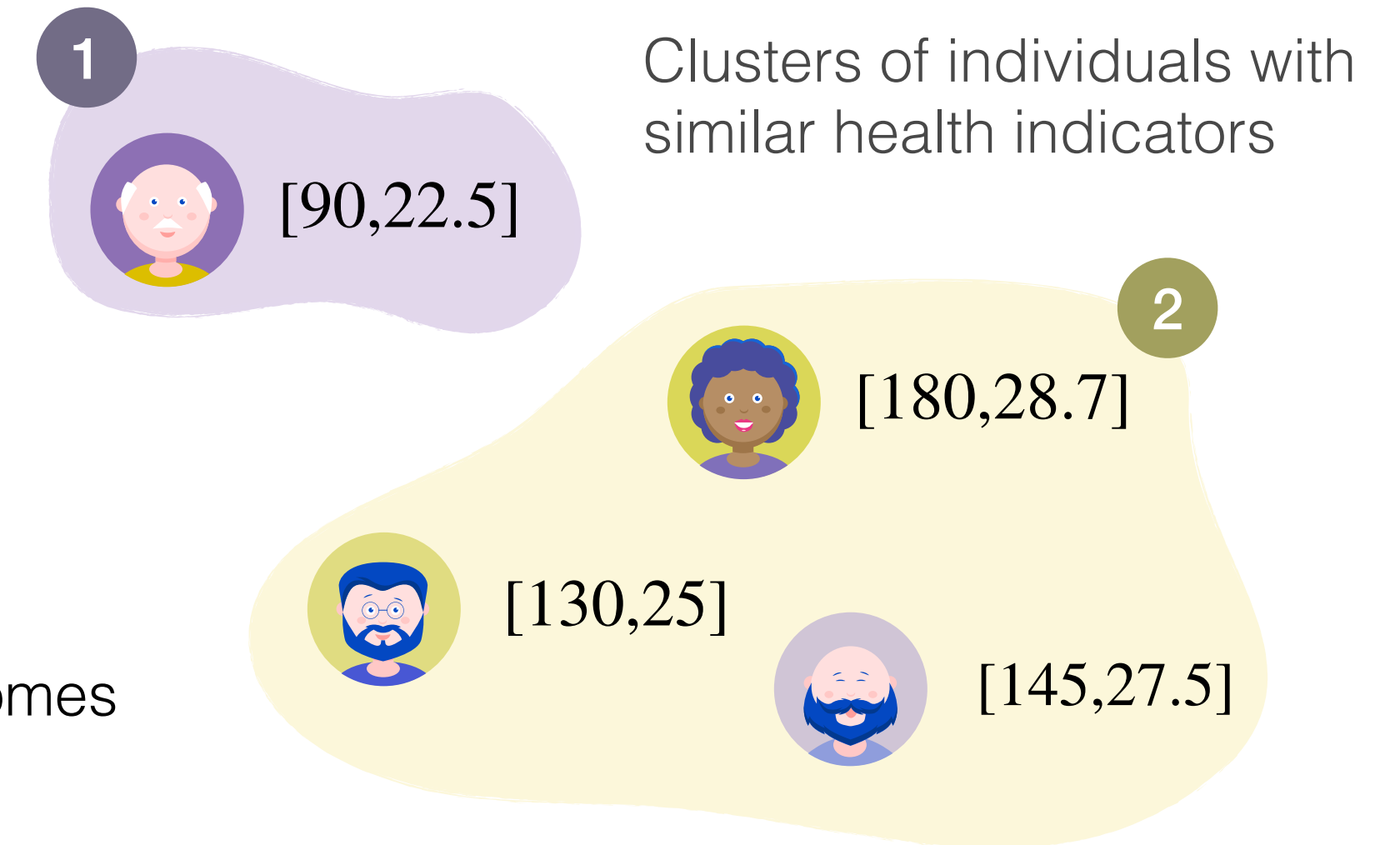


Task– Discover hidden patterns or structures in the data to potentially reveal patterns like diabetes risk levels or other health conditions.

Experience– Suppose we have a dataset of individuals with the following columns:

- ▶ x_1 : Fasting blood sugar level
- ▶ x_2 : Body Mass Index (*BMI*)
- ▶ Unlike supervised learning, unsupervised learning does not use labeled outcomes like diabetes development variable.

Performance– In unsupervised learning, measuring performance is often more challenging due to the absence of a ground truth for comparison. *More on this in future lectures!*



Formal Presentation– We represent our dataset as a collection of pairs: $\{x^{(1)}, x^{(2)}, \dots, x^{(n)}\}$, where:

- ▶ $x^{(i)} = [x_1^{(i)}, x_2^{(i)}]$: is a vector representing the health indicators of the i^{th} individual.

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